

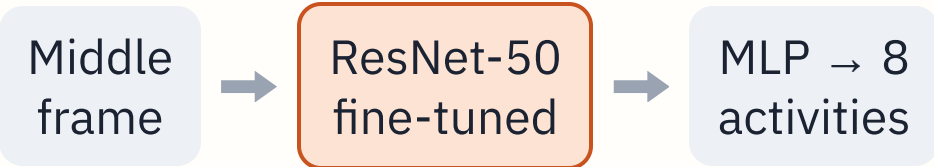
From one frame to players and time

Test accuracy, 1,337 held-out clips
Highlighted = new in this step

B1 · Single frame

62.6%

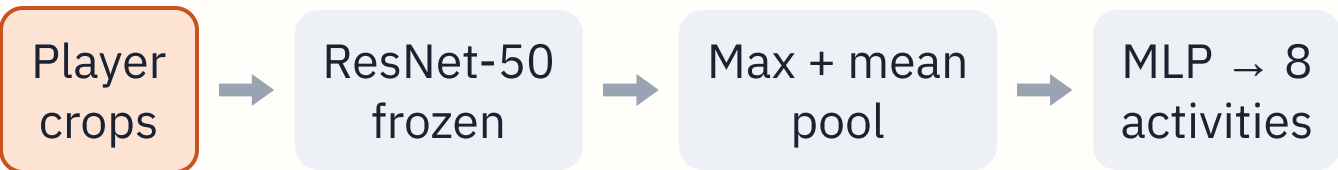
One middle frame, no players, no time. The floor every later model has to beat.



B3 · Player crops

60.7%

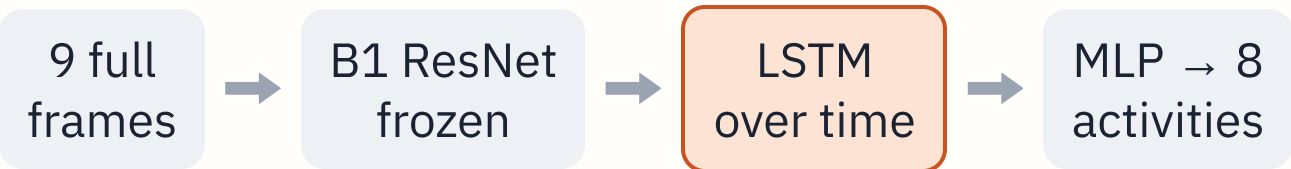
Adds players: a ResNet trained on the 9 player actions, pooled across everyone on court. No time yet.



B4 · Frames over time

66.1%

Adds time instead: features from all 9 frames run through an LSTM. No players.



Players over time, then the full hierarchy

Test accuracy, 1,337 held-out clips
Highlighted = new in this step

B5 · Player LSTM

66.3%

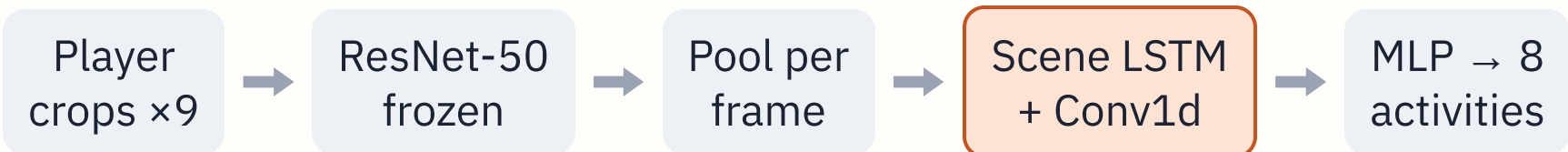
Players and time together: each player's 9 frames go through their own LSTM, then players are pooled.



B6 · Scene LSTM

70.5%

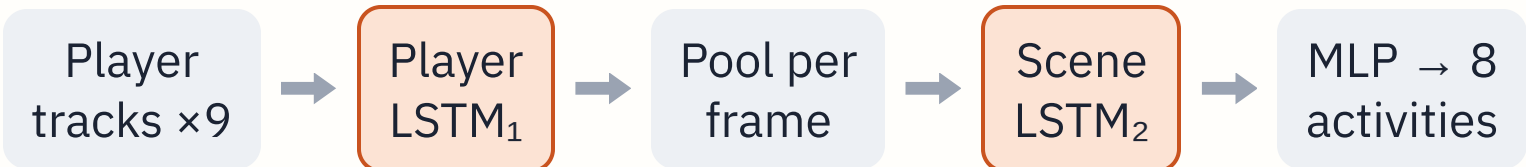
Time at the scene level: pool the players in each frame, then one LSTM over the clip, with a skip connection.



B7 · Hierarchy

73.8%

The paper's model: a player LSTM, pooling per frame, then a scene LSTM, on frozen ResNet-50 features.



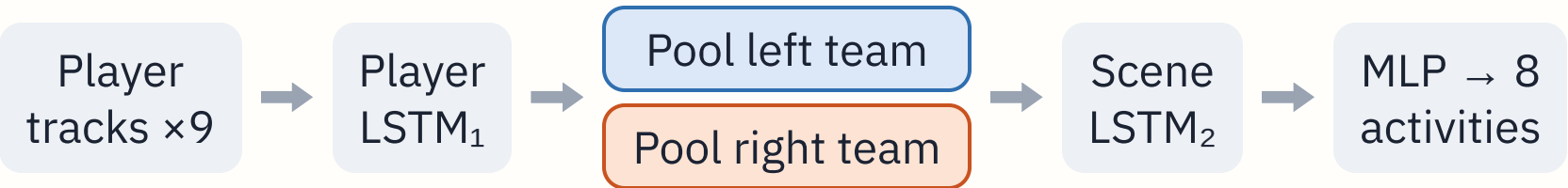
One change wins: pool each team separately

Test accuracy, 1,337 held-out clips
Highlighted = new in this step

B8 · Team split

85.6%

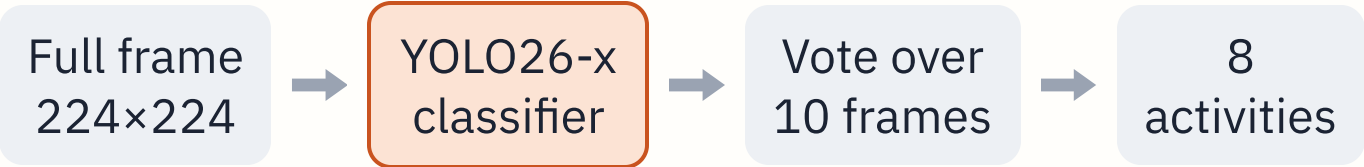
B7 with one change: each team is pooled on its own, so left and right stay apart. +11.9 points over B7.



B9 · Control

80.3%

One YOLO classifier on the whole frame. No boxes, no tracks, no player labels; frames vote per clip.



85.6%

B8 beats the CVPR 2016 paper (81.9%) and stays about 5 points ahead of the frame-only B9, so player structure pays once the teams are kept apart.

Code: github.com/omarTBakr/Volleyball-Group-Activity-Recognition · Demo: huggingface.co/spaces/OmarTBakr/volleyball-activity-demo